

Question 1. Consider $f(x, y) = xy - 2x - y$ and $g(x, y) = 2x + y$.

(a) What is the maximum value of $f(x, y)$ subject to the constraint $g(x, y) = 20$?

(b) If $f(x, y)$ is the profit as a function of minutes of advertising on TV (x) and podcast (y), and $g(x, y)$ is the total cost of advertising (so xy is the revenue). What is the real-world interpretation of the Lagrange multiplier λ ? (Select multiple and argue why they must be equal.)

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|---|--|
| ☐ The profit gained per minute of advertising on TV. | ☐ The profit gained per minute of advertising on podcast. |
| ☐ The profit gained per dollar spent on advertising on TV. | ☐ The profit gained per dollar spent on advertising on podcast. |

Question 2. Find the point on the graph of $y = x^2 + 4x + 6$ that is the closest to the line $y = 2x$.

Question 3. Consider the tilted ellipse $5x^2 - 6xy + 5y^2 = 4$. Find the points on the ellipse that are closest to and farthest from the origin.

Question 4 (Second derivative test). Find the critical points of $f(x, y) = x^2 + 3xy + y^2$ and determine whether they are local maxima, minima, or saddle points.

Question 5. The following boxed arguments are incorrect or incomplete. Identify the error in each argument and provide a correct solution.

Question: Determine whether $\sum_{n=2}^{\infty} \frac{1}{(\ln(n)+2)^2}$ converges.

Flawed Argument: Yes, because as n approaches infinity, $\ln(n) + 2$ approaches infinity, so $\frac{1}{(\ln(n)+2)^2}$ approach zero which is less than one.

Question: Find the interval of convergence of the power series $\sum_{n=0}^{\infty} \frac{(x-2)^n}{5n+3}$.

Flawed argument: Apply ratio test, the series converges when

$$\lim_{n \rightarrow \infty} \left| \frac{(x-2)^{n+1}}{5(n+1)+3} \frac{5n+3}{(x-2)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{5n+3}{5n+8} (x-2) \right| = |x-2| < 1.$$

Thus, $-1 < x-2 < 1$ which is equivalent to $1 < x < 3$, so the interval of convergence is $(1, 3)$.

Question: Calculate the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2}$ or show that it does not exist.

Flawed argument: Along the x -axis, the limit is $\lim_{x \rightarrow 0} \frac{x \cdot 0}{x^2+0} = 0$ and along the y -axis, the limit is $\lim_{y \rightarrow 0} \frac{0 \cdot y}{0+y^2} = 0$, so the limit is 0.

Question: Find the tangent line to the curve $y = y^2 - x^3 + 1$ at the point $(1, 0)$.

Flawed argument: We compute the gradient of $f(x, y) = y^2 - x^3 + 1$ which at the point $(1, 0)$ is

$$\nabla f(x, y)|_{(1,0)} = \langle -3x^2, 2y \rangle|_{(1,0)} = \langle -3, 0 \rangle.$$

The tangent line is the line orthogonal to the gradient and passing through the point $(1, 0)$, so the equation of the line is $\langle x-1, y-0 \rangle \cdot \langle -3, 0 \rangle = 0$ which is simply $x = 1$.

Question: Let $f(x, y) = (x-1)y$ and $x(t) = e^t$ and $y(t) = \cos(y)$. Find the first two nonzero terms of the Maclaurin series of $f(x(t), y(t))$.

Flaws argument: We write $x(t) = 1 + t$ and $y(t) = 1 - \frac{t^2}{2}$, then we get

$$f(x(t), y(t)) = (1+t-1) \left(1 - \frac{x^2}{2} \right) = t - \frac{t^3}{2}.$$